

JEE MAIN CUMULATIVE TEST SERIES

44
YEARS
OF EXCELLENCE



NARAYANA
IIT-JEE ACADEMY-INDIA

Sec: ISR_*CO-SC(MODEL-B)

Date: 07-04-2024

Time: 3 Hrs

CTM-34

Max. Marks: 300

PHYSICS

MAX.MARKS: 100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A piano string 2 m long is made of steel of density $7.7 \times 10^3 \text{ kg/m}^3$ and Young's modulus $2 \times 10^{11} \text{ N/m}^2$. It is maintained at a tension which produces an elastic strain of 1% in the string. The fundamental frequency of transverse vibrations of string is
(A) 85 Hz (B) 170 Hz (C) 340 Hz (D) 127.5 Hz
2. A charge Q is uniformly distributed over a long rod AB of length L as shown in the figure. The electric potential at the point O lying at a distance L from the end A is



(A) $\frac{Q}{8\pi\epsilon_0 L}$

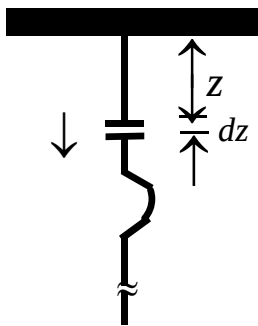
(B) $\frac{3Q}{4\pi\epsilon_0 L}$

(C) $\frac{Q}{4\pi\epsilon_0 L \ln 2}$

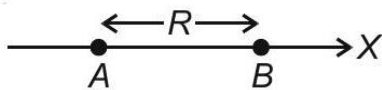
(D) $\frac{Q \ln 2}{4\pi\epsilon_0 L}$

3. A long rope is designed as shown in the figure such a way that v (velocity of the transverse pulse) is independent of z (distance from point of suspension). For such a rope μ cannot be constant, where μ is linear density of rope. Then the general equation for $\mu(z)$ is

[use μ_0 as mass per unit length of the rope at point $z=0$]



- (A) $\mu = 4\mu_0 e^{-(g/v^2)z}$ (B) $\mu = 4\mu_0 e^{(g/v^2)z}$
 (C) $\mu = \mu_0 e^{-(g/v^2)z}$ (D) $\mu = 4\mu_0 e^{(g/v^4)z}$
4. The equation of a stationary wave is $y_{(\text{in cm})} = 4 \sin\left(\frac{\pi x}{5}\right) \cos(100\pi t)$. The wave is formed using a string of length 20 cm. The second and third antinodes are located at positions (in cm)
- (A) 7.5, 12.5 (B) 2.5, 7.5 (C) 12.5, 17.5 (D) 5, 10
5. Two electric dipoles, A, B with respective dipole moments $\vec{p}_A = -4qa\hat{i}$ and $\vec{p}_B = -2qa\hat{i}$ are placed on the x -axis with a separation R , as shown in the figure

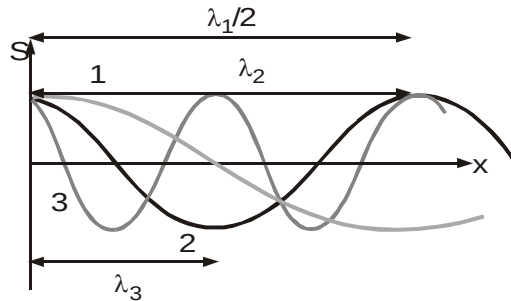


The distance from A at which both of them produce the same potential is

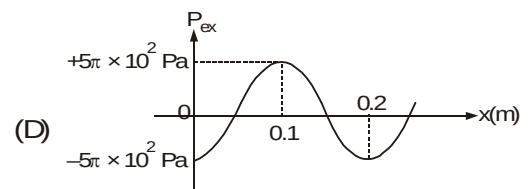
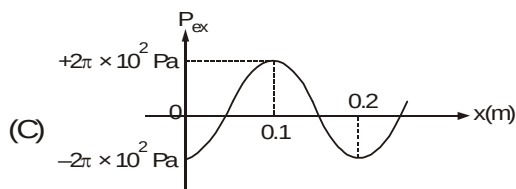
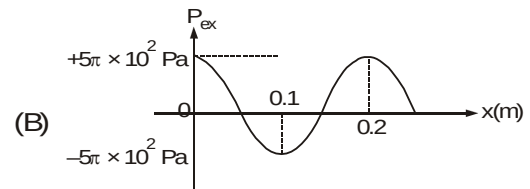
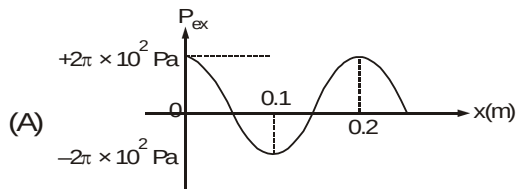
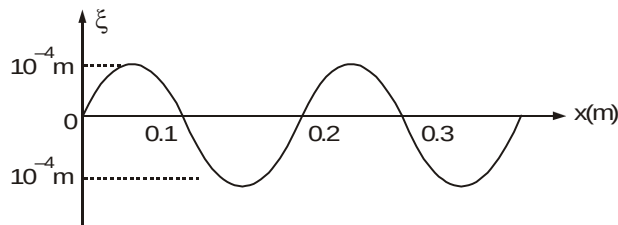
- (A) $\frac{\sqrt{2}R}{\sqrt{2}-1}$ (B) $\frac{\sqrt{2}R}{\sqrt{2}+1}$
 (C) $\frac{R}{\sqrt{2}-1}$ (D) $\frac{R}{\sqrt{2}+1}$

6. Determine the change in volume of 6 litres of alcohol if the pressure is decreased from 200 cm of Hg to 75 cm. [velocity of sound in alcohol is 1280 m/s, density of alcohol = 0.81 gm/cc, density of Hg = 13.6 g/cc and $g = 9.81 \text{ m/s}^2$]
 (A) 0.45 cc (B) 1.25 cc (C) 0.75 cc (D) 1.75 cc
7. A long taut string is connected to a harmonic oscillator of frequency 'f' at one end. The oscillator oscillates with an amplitude ' a_0 ' and delivers power ' p_0 ' to the string. Due to dissipation of energy the amplitude of wave goes on decreasing with distance 'x' from the oscillator given as $a = a_0 e^{-kx}$ where k is constant. At what distance from the oscillator, the power dissipated is equal to 3/4 th of power supplied by the source?
 (A) $\frac{\ln 2}{k}$ (B) $\frac{\ln 2}{2k}$ (C) $\frac{\ln 3}{2k}$ (D) $\frac{\ln 3}{k}$
8. A small steel ball of mass 5 g is dropped from a height of 2 m on a hard floor. If 0.001% of its kinetic energy before striking the floor gets converted into a sound pulse having a duration of 0.4 sec. Find the distance of farthest point from striking point, sound can be heard, if minimum audible intensity is $2 \times 10^{-8} \text{ Wm}^{-2}$. (Assume no loss of sound energy in the atmosphere)
 (A) 1.51 m (B) 2.15 m (C) 3.15 m (D) 4.51 m
9. A plane sinusoidal undiminished sound wave is travelling in +x direction in a gaseous medium. Excess pressure amplitude for any location is P_0 . Wave speed in the medium is 360 ms^{-1} . At some moment, instantaneous values of excess pressures at positions (0, 0, 0) m & (+1, +1, 0) m are $+P_0$ and $+\frac{P_0}{2}$. The possible frequency of wave is:
 (A) 60 Hz (B) 120 Hz (C) $60\sqrt{2}$ Hz (D) $120\sqrt{2}$ Hz
10. Three concentric metal shells A, B and C of respective radii a, b and c ($a < b < c$) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively. The potential of shell B is
 (A) $\frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{a} + c \right]$ (B) $\frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{b} + c \right]$
 (C) $\frac{\sigma}{\epsilon_0} \left[\frac{b^2 - c^2}{b} + a \right]$ (D) $\frac{\sigma}{\epsilon_0} \left[\frac{b^2 - c^2}{c} + a \right]$

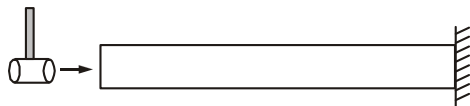
11. Figure shown is a graph, at a certain time t , of the displacement function $S(x, t)$ of three sound waves 1, 2 and 3 as marked on the curves that travel along x -axis through air. If P_1 , P_2 and P_3 represent their pressure amplitudes respectively, then correct relation between them is:



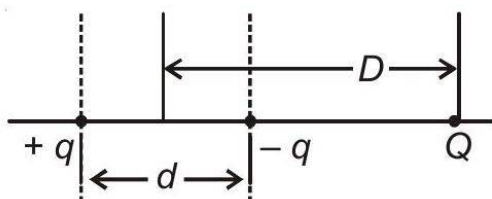
- (A) $P_1 > P_2 > P_3$ (B) $P_3 > P_2 > P_1$ (C) $P_1 = P_2 = P_3$ (D) $P_2 > P_3 > P_1$
12. For a sound wave travelling towards $+x$ direction, sinusoidal longitudinal displacement at a certain time is given as a function of x . If Bulk modulus of air is $B = 5 \times 10^5 \text{ N/m}^2$, the variation of pressure excess will be:



13. A 100 m long rod of density $10.0 \times 10^4 \text{ kg/m}^3$ and having Young's modulus $Y = 4 \times 10^{11} \text{ Pa}$, is clamped at one end. It is hammered at the other free end. The longitudinal pulse goes to right end, gets reflected and again returns to the left end. How much time, the pulse takes to go back to initial point.



- (A) 0.1 s (B) 0.2 s (C) 0.3 s (D) 2 s
14. A sound wave of wavelength 40 cm travels in air. If the difference between the maximum and minimum pressures at a given point is $2.0 \times 10^{-3} \text{ N/m}^2$. The amplitude of vibration of the particles of the medium is (The bulk modulus of air $2 \times 10^5 \text{ N/m}^2$)
- (A) $1.185 \times 10^{-10} \text{ m}$ (B) $2.27 \times 10^{-10} \text{ m}$
 (C) $4.185 \times 10^{-10} \text{ m}$ (D) $3.185 \times 10^{-10} \text{ m}$
15. The faintest sound, the human ear can detect at a frequency of 1 kHz (for which ear is most sensitive) correspond to an intensity of about 10^{-12} W/m^2 (the so called threshold of hearing). The maximum displacement (amplitude) associated with this sound wave is (assume the density of air = 1.3 kg/m^3 and velocity of sound in air = 332 m/s)
- (A) $4.085 \times 10^{-11} \text{ m}$ (B) $3.085 \times 10^{-11} \text{ m}$ (C) $1.085 \times 10^{-11} \text{ m}$ (D) $2.085 \times 10^{-11} \text{ m}$
16. A closed organ pipe of radius r_1 and an open organ pipe of radius r_2 and having same length 'L' resonate when excited with a given tuning fork. Closed organ pipe and open organ pipe resonates in fundamental mode, then
- (A) $1.2 (r_2 - r_1) = L$ (B) $r_2 - r_1 = L$ (C) $r_2 - 2r_1 = 2.5L$ (D) $2r_2 - 2r_1 = 2.5L$
17. A system of three charges are placed as shown in the figure



If $D > d$, the potential energy of the system is best given by

(A) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right]$

(B) $\frac{1}{4\pi\epsilon_0} \left[+\frac{q^2}{d} + \frac{qQd}{D^2} \right]$

(C) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} + \frac{2qQd}{D^2} \right]$

(D) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{2D^2} \right]$

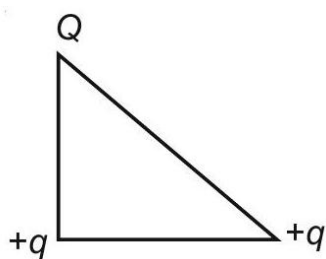
18. A granite rod of 60 cm length is clamped at its middle point and is set into longitudinal vibrations. The density of granite is $2.7 \times 10^3 \text{ kg/m}^3$ and its Young's modulus is $9.27 \times 10^{10} \text{ Pa}$. What will be the fundamental frequency of the longitudinal vibrations?
- (A) 5 k Hz (B) 2.5 k Hz (C) 10 k Hz (D) 7.5 k Hz
19. A resonance tube is old and has jagged end. It is still used in the laboratory to determine velocity of sound in air. A tuning fork of frequency 512 Hz produces first resonance when the tube is filled with water to a mark 11 cm below a reference mark, near the open end of the tube. The experiment is repeated with another fork of frequency 256 Hz which produces first resonance when water reaches a mark 27 cm below the reference mark. The velocity of sound in air, obtained in the experiment, is close to
- (A) 322 ms^{-1} (B) 341 ms^{-1} (C) 328 ms^{-1} (D) 335 ms^{-1}
20. A string 2.0 m long and fixed at its ends is driven by a 240 Hz vibrator. The string vibrates in its third harmonic mode. The speed of the wave and its fundamental frequency are respectively,
- (A) 320 m/s, 120 Hz (B) 320 m/s, 80 Hz
- (C) 180 m/s, 80 Hz (D) 180 m/s, 120 Hz

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

21. In a closed organ pipe, the frequency of fundamental note is 30 Hz. A certain amount of water is now poured in the organ pipe so that the fundamental frequency is increased to 110 Hz. If the organ pipe has a cross-sectional area of 2 cm^2 , the amount of water poured in the organ tube is _____ g.
(Take speed of sound in air is 330 m/s)
22. What is the maximum possible sound level in dB of sound waves in air. Given that density of air = 1.3 kg/m^3 , $v = 332 \text{ m/s}$ and atmospheric pressure $P = 1.01 \times 10^5 \text{ N/m}^2$.
23. Three charges $Q, +q$ and $+q$ are placed at the vertices of a right-angle isosceles triangles as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is $\frac{-\sqrt{x} q}{\sqrt{2}+1}$. The value of x is



24. A transverse wave propagating on a stretched string of linear density $3 \times 10^{-4} \text{ kg/m}$ is represented by the equation $y = 0.2 \sin(1.5x + 60t)$ where 'x' is in meters and 't' is in seconds. The tension in string (in newton) is

25. A steel wire of length 1 m, mass 0.1 kg and uniform cross sectional area 10^{-6} m^2 is rigidly fixed at both ends. The temperature of the wire is lowered by 20°C . If the transverse waves are set up plucking the string in the middle, calculate the frequency (in Hz) of the fundamental mode of vibration. ($Y = 2 \times 10^{11} \text{ N/m}^2$, $\alpha = 1.21 \times 10^{-5}/^\circ\text{C}$)
26. A plane longitudinal wave having angular frequency $\omega = 500 \text{ rad/sec}$ is travelling in positive x- direction in a medium of density $\rho = 1 \text{ kg/m}^3$ and bulk modulus $4 \times 10^4 \text{ N/m}^2$. The loudness at a point in the medium is observed to be 20 dB. Assuming at $t = 0$ initial phase of the medium particles to be zero. The maximum pressure change in the medium is $N \times 10^{-4} \text{ Nm}^{-2}$. The value of N is
27. A close organ pipe diameter 10 cm has length 42 cm. The air column in pipe vibrates in its second overtone with the maximum amplitude ΔP_0 . The pressure amplitude at middle of pipe is $\frac{\sqrt{x} \Delta P_0}{2}$. The value of x is
28. A uniform string of length l , fixed at both ends is vibrating in its 2nd overtone. The maximum amplitude is 'a' and tension in string is 'T'. If the energy of vibration contained between two consecutive nodes is $\frac{K a^2 \pi^2 T}{8 l}$, then the value of 'K' is
29. An electric field $\vec{E} = \vec{i}Ax$ exist in the space, where $A = 10 \text{ V/m}^2$. Take the potential at $(10\text{m}, 20\text{m})$ to be zero. The potential (in volt) at the origin, is
30. A man standing in front of a mountain at a certain distance beats a drum at regular intervals. The drumming rate is gradually increased and he finds that the echo is not heard distinctly when the rate becomes 40 per minute. He then moves nearer to the mountain by 80 metre and finds that the echo is again not heard when the drumming rate become 60 per minute. The velocity (in m/s) of sound is

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Surfactant molecules can cluster together as micelles, which are colloid sized cluster of molecules. Micelles form only above critical micelle concentration (CMC) and above certain temperature called Kraft temperature. ΔH of micelle formation can be positive or negative. Which is false about micelle formation?
- (A) ΔS of micelle formation is positive
(B) the hydrophobic part lie towards interior of micelle
(C) the hydrophilic part lie towards surface of micelle
(D) ΔS of micelle formation is negative
32. Among the following gases, CO_2 , SO_2 , CH_4 and H_2 which of the options is correct w.r.t. physisorption on a solid surface :
- (A) The extent of adsorption will be maximum in case of H_2 , least for CO_2 .
(B) The extent of adsorption will be same for all the above gases.
(C) The extent of adsorption maximum for SO_2 and minimum for H_2 (g)
(D) All the options are incorrect.
33. Which of the following statements is correct with respect to Langmuir adsorption theory?
- [$x \rightarrow$ represents mass of adsorbed gas, $m \rightarrow$ represents mass of adsorbent]
- (A) Graph of $\frac{x}{m}$ vs P will be linear always.
(B) Graph of $\frac{m}{x}$ vs P will be linear always.
(C) Graph of $\frac{m}{x}$ vs $\frac{1}{P}$ will be linear always.

(D) Graph of $\frac{x}{m}$ vs $\frac{1}{P}$ will be linear always.

34. Identify the option which is not correct :

(A) Physical adsorption is multilayer, non-directional and nonspecific.

(B) As a result of adsorption, there is an increase of surface energy.

(C) In case of adsorption in solution, it is possible that solvent molecules may get absorbed preferentially as compared to solute molecules.

(D) A crystalloid and a colloid differs in their ability to show Tyndall effect.

35. One gm of charcoal adsorbs CH_3COOH from 100 ml 0.6 M CH_3COOH aqueous solution to form a monolayer, and thereby the molarity of CH_3COOH reduces to 0.58. Calculate the surface area of the charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = $3.0 \times 10^2 \text{ m}^2/\text{gm}$.

(A) 5 nm^2 (B) 0.5 nm^2 (C) 0.25 nm^2 (D) 2.5 nm^2

36. For a process of adsorption of a gas on metal (solid) surface, known as occlusion, the enthalpy change is about -35 Kcal/mol at 25°C and 1bar. The only correct statement regarding adsorption is:

(A) on increasing the temperature, the extent of adsorption of gas will decrease.

(B) the adsorption is reversible.

(C) the activation energy of adsorption may be nearly zero.

(D) adsorption is monolayer.

37. When blood containing "Haemoglobin" as the colloidal partial is subjected to electro osmosis. The dispersion medium moves towards the anode. Hence, the incorrect statement will be

(A) $\text{Fe}(\text{OH})_3$ solution on mixing with haemoglobin in blood can cause coagulation

(B) As_2S_3 solution can cause coagulation in blood on mixing.

(C) In electrophoresis of blood, movement of colloidal particle will be towards cathode.

(D) NaCl on mixing in blood can cause coagulation.

38. Which of the following options correctly represent set of True (T) or False (F) statements.
- (I) Colloidal solution of gelatin gets stabilised by solvation of colloidal particles by the dispersion medium.
 - (II) An aqueous solution of KI will not show Tyndall effect, however when added with excess AgNO_3 solution it will show Tyndall effect.
 - (III) At isoelectric point all the colloids will have least stability
 - (IV) In an autocatalysis reaction, rate of reaction will increase with time.
 - (V) Physisorption is an enthalpy driven process.
- (A) TTFTT (B) FTTTF (C) TFTFT (D) FTFFF
39. Identify the statements which is incorrect w.r.t. surface phenomenon.
- (A) A sol, prepared by adding excess AgNO_3 solution in KI solution when subjected to electro-phoresis will show coagulation near cathode.
 - (B) The conductivity of a soap solution decreases steeply at CMC.
 - (C) Tyndall effect will be more prominent in gold solution as compared to rubber solution.
 - (D) Extent of physisorption increases with increase in temperature initial and then decreases with temperature.
40. If $\text{Fe}(\text{OH})_3$ sol is subjected to electrophoresis then which of the following statement regarding coagulation will be correct.
- (A) No coagulation will occur at any electrode.
 - (B) Coagulation will occur at cathode
 - (C) Coagulation will occur at anode
 - (D) Coagulation will occur at both the electrodes

41. For a unielectronic specie, compare the energy of the following orbitals (represented by S)
- S_1 = a spherically symmetrical orbital having two spherical nodes.
 S_2 = an orbital which is double - dumb bell and has no radial nodes.
 S_3 = an orbital with orbital angular momentum zero and three radial nodes.
 S_4 = an orbital having one planar and one radial node.
- (A) $S_1 = S_2 = S_3 = S_4$ (B) $S_1 = S_2 = S_4 < S_3$
(C) $S_1 > S_2 > S_3 > S_4$ (D) $S_1 < S_2 < S_3 < S_4$
42. An aqueous solution of sodium stearate, $C_{17}H_{35}COONa$, becomes a colloidal sol at the concentration of 1.2×10^{-3} M at $25^\circ C$. If at this concentration, 2.4×10^{13} colloidal (micelle) particles are present per mm^3 , the number of stearate ions in each colloidal particle is : ($N_A = 6 \times 10^{23}$)
- (A) 30 (B) 60
(C) 90 (D) 120
43. Give the correct order of initials T (true) or F (false) for following statements.
- (i) The ratio of the radii of the first three Bohr orbit of hydrogen atom is 1 : 8 :
(ii) The ratio of magnitude of Total Energy : Kinetic Energy : Potential Energy for any orbit is 1 : 1 : 2
(iii) The frequency of a green light is 6×10^{14} Hz, then its wavelength is 500 nm
(iv) The ratio of de-broglie wavelength of a 'H' atom, 'He' atom and CH_4 molecule moving with equal kinetic energy is 4 : 2 : 1
- (A) FFTT (B) FTFT (C) TFTF (D) FTTT
44. Which of the following is/are correct ?
- (A) Tanning is an example of mutual coagulation.
(B) Argyrol sol is gold sol.
(C) When 200 ml of 1M $AgNO_3$ is mixed with 500 ml of 0.5 M KI, resulting sol formed will be positively charged.

(D) On mixing small amount of lyophilic colloid to lyophobic colloid, later can be destabilized

45. Identify incorrect statement :

(A) During cataphoresis gold sol will move towards anode.

(B) A starch aqua sol can act as protective colloid for sulphur sol.

(C) In chemisorption extent of adsorption increases on increasing temperature at low temperatures.

(D) For Tyndall effect to be observed refractive index of dispersed phase and dispersion medium should be almost same.

46. Which of the following statement is incorrect?

(A) The elevation in boiling point of an alcoholic solution of sulphur is more than that of its sol in water, if mass of sulphur present per unit volume of mixture is same in both case.

(B) CMC value of $\text{CH}_3(\text{CH}_2)_9\text{NH}_2\text{Cl}$ will be less than that of $\text{CH}_3(\text{CH}_2)_6\text{COONa}$.

(C) Freundlich adsorption theory is followed at all pressures.

(D) $\text{CO}_2(\text{g})$ can displace adsorbed $\text{O}_2(\text{g})$ from surface of an adsorbent showing physisorption.

47. Which of the following will be incorrect statement -

(A) Tyndall effect is used to distinguish between a colloidal and true solution.

(B) In the coagulation of a negative sol, the flocculating power is in the order : $\text{Al}^{3+} > \text{Ba}^{2+} > \text{Na}^+$

(C) When river water meets the sea water, the electrolytes present in sea water coagulate the colloidal solution of clay resulting in its deposition with the formation of Delta.

(D) In the coagulation of a positive sol, the flocculating power is in the order: $\text{SO}_4^{2-} > \text{PO}_4^{3-} > [\text{Fe}(\text{CN})_6]^{4-}$.

48. Identify the correct statement :
- (A) Lyophobic colloid is used as protective colloid.
 (B) Emulsion can be diluted with any amount of dispersed phase.
 (C) During electrophoresis clay will accumulate at anode.
 (D) Foam rubber is gas in liquid.
49. Calculate the maximum and minimum number of electrons. Which may have magnetic quantum number $m = +1$ and spin quantum number $s = +\frac{1}{2}$ in Chromium (Cr).
- (A) 3, 2 (B) 6, 4 (C) 4, 2 (D) 2, 1
50. The ratio of magnetic moment and number of pairs of electron present in Co^{+3} is
- (A) $\frac{\sqrt{6}}{5}$ (B) $\frac{\sqrt{25}}{10}$ (C) $\frac{2\sqrt{6}}{9}$ (D) $\frac{2\sqrt{5}}{10}$

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

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51. Calculate coagulation value of AlCl_3 if for coagulation of 400 ml of a gold sol, 427.2 milli-grams of AlCl_3 is required.
52. If the radius of first Bohr orbit of He^+ ions is $x \text{ cm}$ and radius of 4th Bohr orbit of Be^{+3} is $y \text{ cm}$, then calculate the value of y/x .
53. For adsorption of gas over solid surface following data is obtained at 300K.

Pressure of gas (mm of Hg)	100	25
Amount adsorbed per Kg of charcoal	3 gm	1.5 gm

The slope of the graph between $\log P$ vs $\log x/m$ will be :

[x/m and P are in same units as given in question.]

54. 6.84 gm $\text{Al}_2(\text{SO}_4)_3$ is needed to coagulate 2.5L of As_2S_3 sol completely in 2.0 hrs. The coagulation value of $\text{Al}_2(\text{SO}_4)_3$ is :
55. If the radius of first Bohr orbit of He^+ ions is **x cm** and radius of 4th Bohr orbit of Be^{+3} is **y cm**, then calculate the value of y/x.
56. For the coagulation of 100 mL of arsenious sulphide solution, 5 mL of 0.1 M NaCl was used. What is the flocculation value of NaCl?
57. In order to cause coagulation of 100 ml Arsenious sulphide sol 111.7 mg of 2 m NaCl solution is used what will be the flocculation value of NaCl ?
58. At a temperature of 180 K, O_2 gas adsorbs on Pt surface following Freundlich adsorption isotherm. Following experimental data is obtained.

P (bar)	2.5	10	81
x / m	0.15	0.3	0.8538

where P is partial pressure of gas & $\frac{x}{m}$ is mass of gas adsorbed per gm of platinum.

Calculate millimoles of O_2 adsorbed per gm of platinum at 4 bar pressure & 180 K.

59. In order to coagulate 400 ml of $\text{Fe}(\text{OH})_3$ sol completely 60 ml of 0.1M KCl is required. What will be the coagulation power of KCl?
60. Calculate the number of sols which are negatively charged.

$\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ sol

Sb_2S_3 sol

Ag sol

Cu sol

Blood

basic dye

sol of clay

sol of starch

sol of sodium stearate

sol of sodium laurylsulphate

Acidic dye

sol of Aluminium hydroxide

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. If $P(3,1), Q(6,5)$ and $R(x,y)$ are three points such that ΔPQR is right angled triangle, right angled at R and the area of ΔPQR is 7 square units then the number of such points R is _____
 (A) 0 (B) 2 (C) 4 (D) Infinite
62. In triangle ABC, equation of side BC is $x-y=0$. Circumcentre and orthocenter of the triangle are (2,3) and (5,8) respectively. Equation of circumcircle of the triangle is:
 (A) $x^2 + y^2 - 4x + 6y - 40 = 0$ (B) $x^2 + y^2 - 4x - 6y - 27 = 0$
 (C) $x^2 + y^2 + 4x + 6y - 27 = 0$ (D) $x^2 + y^2 + 4x - 6y - 27 = 0$
63. AB is a chord of the circle $x^2 + y^2 = 9$. The tangents at A and B intersect at C. If (1,2) is the midpoint of AB, the area of triangle ABC
 (A) 5 (B) 8 (C) $\frac{2\sqrt{2}}{\sqrt{5}}$ (D) $\frac{8}{\sqrt{5}}$
64. The ends of the latus rectum of the conic $x^2 + 10x - 16y + 25 = 0$ are
 (A) (3, -4), (13, 4) (B) (-3, -4), (13, -4)
 (C) (3, 4), (-13, 4) (D) (5, -8), (-5, 8)
65. If the focus of a parabola is (-2,1) and the directrix has the equation $x + y = 3$, then the vertex is:
 (A) (0,3) (B) $\left(-1, \frac{1}{2}\right)$ (C) (-1,2) (D) (2,-1)
66. The coordinates of the focus of the parabola $y^2 = 4(x + y)$, are
 (A) (-1, 1) (B) (0, 2) (C) (2, 1) (D) (2, -1)

67. If $\frac{x^2}{f(4a)} + \frac{y^2}{f(a^2-5)} = 1$ represents an ellipse with major axis is y-axis and f is a positive decreasing function, then
 (A) $a \in (-\infty, 1)$ (B) $a \in (5, \infty)$ (C) $a \in (1, 4)$ (D) $a \in (-1, 5)$
68. The length of the longest chord of the ellipse $4x^2 + 9y^2 = 1$ drawn through $\left(0, -\frac{1}{3}\right)$ is $\frac{3}{\sqrt{\lambda}}$ where λ is equal to ____
 (A) 15 (B) 25 (C) 20 (D) 36
69. If a variable tangent of the circle $x^2 + y^2 = 1$ intersect the ellipse $x^2 + 2y^2 = 4$ at P and Q then the locus of the points of intersection of the tangents at P and Q is
 (A) a circle of radius 2 units
 (B) a parabola with focus as (2, 3)
 (C) an ellipse with eccentricity $\frac{\sqrt{3}}{4}$
 (D) an ellipse with length of latus rectum is 2 units
70. For the hyperbola $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$ where the value of α changes, which of the following remain constant?
 (A) Abscissae of the vertices (C) Abscissae of the foci
 (B) Eccentricity (D) Directrices
71. If the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ & the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide then the value of b^2 is
 (A) 4 (B) 9 (C) 16 (D) none
72. The eccentricity of the hyperbola $9x^2 - 16y^2 - 72x + 96y - 144 = 0$ is
 (A) $\frac{3}{2}$ (B) $\frac{4}{3}$ (C) $\frac{6}{5}$ (D) $\frac{5}{4}$

73. A hyperbola has its axes along the coordinate axes, length of latus rectum is 8 and conjugate axis is half of the distance between the foci. Then the eccentricity is
 (A) $\frac{2}{\sqrt{3}}$ (B) $\frac{3}{\sqrt{2}}$ (C) $\frac{4}{\sqrt{3}}$ (D) $\frac{4}{3}$
74. The distance between two parallel tangents of the hyperbola $\frac{x^2}{9} - \frac{y^2}{49} = 1$ is 2. Then their slopes are
 (A) $\pm \frac{5}{2}$ (B) $\pm \frac{7}{2}$ (C) $\pm \frac{4}{5}$ (D) $\pm \frac{5}{4}$
75. If the tangent to the ellipse $x^2 + 4y^2 = 16$ at the point ' α ' is a normal to the circle $x^2 + y^2 - 8x - 4y = 0$ then $\alpha =$
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $-\frac{\pi}{4}$
76. The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and having its center at (0, 3) is
 (A) 3 (B) 4 (C) $\sqrt{7}$ (D) $\sqrt{12}$
77. If $ax + by + c = 0$ is normal to the curve $y = \frac{x}{1+|x|}$ then
 (A) $ab > 0$ (B) $ab < 0$ (C) $|a| \leq |b|$ (D) $|a| \leq |b|$
78. If the tangent at (1,1) on the curve $y^2 = x(2-x)^2$ meets the curve again at P then P=
 (A) (4,4) (B) (-1,2) (C) (9/4, 3/8) (D) (1,-1)
79. If $y = ax^{n+1} + bx^{-n}$, then $x^2 \frac{d^2y}{dx^2}$ is equal to
 (A) $n(n-1)y$ (B) $n(n+1)y$ (C) ny (D) n^2y
80. If $5f(x) + 3f\left(\frac{1}{x}\right) = x + 2$ and $y = xf(x)$ Then $\left(\frac{dy}{dx}\right)_{x=1}$ equal to
 (A) $\frac{1}{8}$ (B) $\frac{4}{8}$ (C) 2 (D) $\frac{7}{8}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. If centre of the circle given by $x^2 + 5x + y^2 = 9$ lies on a line with slope 2 is $2x + ky + 6 = 0$ then $b = ?$
82. If one diameter of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord to the circle "S" of centre (2,1) then radius of circle S is
83. A line 'L' with slope 2 cuts parabola $y^2 = 8x$ to form a chord. If the midpoint of chord lies on $x = 4$ then the equation of L is $y = 2x + k$. Then $\frac{|k|}{3} = \dots\dots\dots$
84. If AFB is a focal chord of the parabola $y^2 = 4ax$ and $AF = 3$, $FB = 6$, then the latus-rectum of the parabola is equal to
85. The number of values of c such that the straight line $y = 4x + c$ touches the curve $x^2 / 4 + y^2 = 1$ is K then $K = \dots\dots\dots$
86. If $\phi_1, \phi_2, \phi_3, \phi_4$ are the eccentric angles of four concyclic points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $\cos(\phi_1 + \phi_2 + \phi_3 + \phi_4)$ is equal to _____
87. If a variable tangent of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, cuts the circle $x^2 + y^2 = 4$ at point A, B and locus of midpoint of AB is $9x^2 - 4y^2 - \lambda(x^2 + y^2)^2 = 0$ then λ is
88. Distance between the foci of the hyperbola $x^2 - 3y^2 - 4x - 6y - 11 = 0$ is _____
89. A function $f : R \rightarrow R$ satisfies the equation $f(x + y) = f(x) \cdot f(y)$ suppose that the function is differentiable at $x = 0$ and $f'(0) = 2$ also $f'(x) = kf'(x)$ then k^2 is _____
90. Let $f(x) = x + \cos x + 2$ and $g(x)$ the inverse function of $f(x)$, then $g'(3)$ is equal to _____

